WHEN MORE DATA BARELY HURTS: A NEGATIVE RESULT ON DISTRIBUTIONAL OVER-SMOOTHING

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ABSTRACT

Deep models deployed over time encounter new input distributions, and continual learning methods are usually evaluated by how well they prevent catastrophic forgetting. We investigated a related but more geometric failure mode that we call *distributional oversmoothing*: the possibility that training on additional shifted distributions makes representations or decision boundaries too shared across domains, erasing distribution-specific discriminative structure. Our original hypothesis predicted that more diverse sequential data would substantially hurt earlier distributions even with replay. In controlled rotation experiments on sklearn digits, MNIST, and Fashion-MNIST resized to 8×8, this strong hypothesis was not supported. With robust normalization and 50% replay, final original-distribution accuracy dropped by only 2.4 percentage points on sklearn digits, improved slightly on MNIST, and was essentially unchanged on Fashion-MNIST; our proposed Distributional Oversmoothing Gap stayed between 1.2 and 2.7 percentage points. The negative result is informative: the apparent risk was sensitive to mundane preprocessing details, especially normalization of rotated border pixels, and simple replay was enough to make the phenomenon small in these benchmarks. We report the diagnostic metrics and limitations so future work can test distributional oversmoothing in harder, higher-resolution, and more natural continual-learning settings.

1 INTRODUCTION

A common intuition in machine learning is that more data, especially more diverse data, should improve generalization. Continual learning complicates this intuition: models trained sequentially may lose performance on previous tasks, a problem usually framed as catastrophic forgetting (**??**). We began from a different concern. Even if forgetting is partially controlled, new distributions might reshape the representation so that class boundaries become overly smooth across domains. In that case the model would not merely forget old examples; it would learn a representation geometry that is less discriminative for distribution-specific features.

We call this hypothesized failure mode *distributional oversmoothing*. The intended distinction from standard forgetting is operational: forgetting measures performance loss on previous data, whereas oversmoothing should be accompanied by reduced class separation or increased cross-distribution confusion despite retaining access to replayed examples. This distinction matters for deployed classifiers in changing environments, where a model may continuously receive shifted data but still be expected to maintain sharp distinctions in the original operating regime.

This paper is a negative and cautionary report. We implemented a small controlled testbed with sequential rotation shifts and replay. The strongest version of the hypothesis did not materialize. Across three datasets, the final original-distribution degradation was small or absent, and representation-separation diagnostics did not reveal a collapse. The main contributions are therefore modest but, we believe, useful for the ICBINB setting: we formalize two lightweight diagnostics for the proposed failure mode; we show that in these small benchmarks robust normalization and replay largely prevent the predicted degradation; and we identify a preprocessing pitfall that can masquerade as a continual-learning pathology.

2 RELATED WORK

Continual learning studies the stability-plasticity trade-off: a learner must adapt to new data while preserving useful previous knowledge (**?**). Regularization-based approaches such as elastic weight consolidation penalize movement in parameters believed to be important for previous tasks; **?** analyzes quadratic penalties used in this family. Replay-based methods instead store or regenerate previous examples and mix them into later updates, often providing strong practical baselines (**??**). Our experiments use replay not as a novel method, but as a control: if a degradation remains under substantial replay, it is less easily dismissed as ordinary forgetting.

Our hypothesis is also related to negative transfer, where source knowledge can harm target performance (**??**). The emphasis here differs: we ask whether sequentially added distributions harm *earlier* domains by changing representation geometry, not whether an old source helps or hurts a new target. Representation comparison tools such as CKA (**?**) motivate geometry-based analysis, although our completed experiments use a simpler centroid-ratio diagnostic rather than CKA. Finally, decision-boundary sharpness and interpretability have long been discussed in both classical and black-box models (**??**); our diagnostic framing borrows this concern but evaluates it through class separation and confusion under distribution shift.

3 METHOD

We consider a sequence of distributions $\mathcal{D}_1, \ldots, \mathcal{D}_T$ over the same label space. The model first trains on the original distribution and then sees rotated variants. Let $a_t(d)$ be validation accuracy on distribution d after epoch t, and let $b_t(d) = \max_{\tau \le t} a_\tau(d)$ be the best accuracy observed so far. We report two accuracy-based diagnostics. The Original Distribution Accuracy Drop is

$$\mathrm{ODAD}_t = a_{\mathrm{anchor}}(\mathcal{D}_1) - a_t(\mathcal{D}_1), \tag{1}$$

where a_{anchor} is the best original-distribution accuracy reached before any shift. We also define the Distributional Oversmoothing Gap,

$$\mathrm{DOG}_t = \frac{1}{|\mathcal{P}_t|} \sum_{d \in \mathcal{P}_t} \max(0, b_t(d) - a_t(d)), \tag{2}$$

where $\mathcal{P}_t$ is the set of previously introduced distributions. DOG is not proof of oversmoothing; it is a screening statistic for losses on prior distributions after new domains are introduced.

To probe representation geometry, we compute a centroid ratio on the final hidden-layer features z_i of original-distribution validation examples:

$$R = \frac{\mathrm{mean}_{c<c'} \|\mu_c - \mu_{c'}\|_2}{\mathrm{mean}_c \mathrm{mean}_{i:y_i=c} \|z_i - \mu_c\|_2 + \epsilon}. \tag{3}$$

Lower values indicate weaker class separation relative to within-class spread. Our initial plan included CKA and explicit boundary-sharpness metrics, but those analyses were not completed in the reported runs; we therefore do not use them to support any claim.

4 EXPERIMENTAL SETUP

We used three classification testbeds: sklearn digits, MNIST, and Fashion-MNIST. All images were represented as 8×8 grayscale inputs so the same architecture could be used throughout. Each dataset was transformed into a sequence of three distributions: original images, images rotated by $+20^\circ$ with small Gaussian noise, and images rotated by -20° with larger Gaussian noise. The sklearn split used 70% training and 30% validation. MNIST and Fashion-MNIST used stratified subsets of at most 6000 training and 2000 validation examples.

A key implementation detail was normalization. An earlier version normalized rotated images using per-pixel standard deviations estimated from the original distribution. Border pixels in the original 8×8 images can have near-zero variance, causing rotated edge pixels to be scaled to extreme values. The reported runs use a standard-deviation floor of 0.10. This mundane fix substantially changed the stability of the experiment and is one reason we treat the results as a cautionary negative finding.

Table 1: Main results. "Anchor" is the best original-distribution accuracy before shifts. ODAD is anchor minus final original accuracy, so negative means improvement. DOG is the final mean best-to-current drop over previously learned distributions. The strong oversmoothing hypothesis would predict large positive ODAD and DOG despite replay; we observe only small gaps.

Dataset	Anchor	Final orig.	Final $+20^\circ$	Final -20°+noise	ODAD	DOG
sklearn digits	0.9796	0.9556	0.9426	0.9611	0.0241	0.0269
MNIST	0.9340	0.9395	0.8950	0.9220	-0.0055	0.0183
Fashion-MNIST	0.8215	0.8210	0.7695	0.7855	0.0005	0.0123

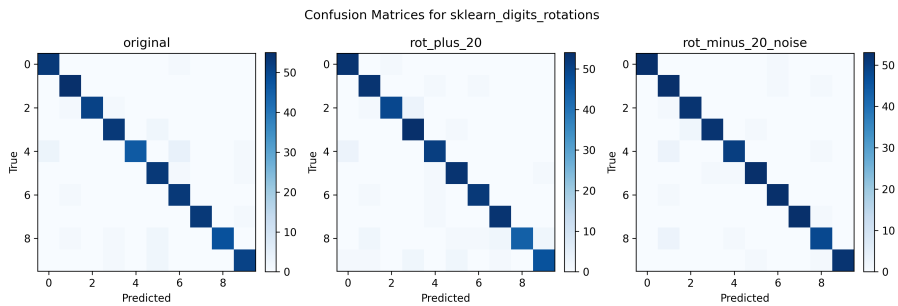

Figure 1: Final sklearn digits confusion matrices after the complete continual-learning sequence. This is the run with the largest observed ODAD. The original-domain matrix remains strongly diagonal despite the final accuracy drop, and shifted matrices show localized off-diagonal errors rather than diffuse cross-class confusion. Rows are true classes and columns are predicted classes; darker cells denote larger counts.

The model is a two-hidden-layer MLP with 128 ReLU units per layer and a linear classifier. We train with AdamW, gradient clipping at norm 5, and experience replay. During each continual phase, the current distribution is mixed with a replay sample from learned distributions using replay fraction 0.50. Sklearn digits used 30 initial epochs and 12 epochs per later distribution; MNIST and Fashion-MNIST used 14 initial epochs and 7 epochs per later distribution. Hyperparameters are listed in Section A.

5 Experiments

Table 1 is the main empirical summary. The strongest evidence for degradation is on sklearn digits, where original-distribution accuracy falls from 0.9796 to 0.9556 after the two shifted distributions. This is a real drop, but it is only 2.4 percentage points and occurs while final accuracy remains high on all three distributions. MNIST does not show original-domain harm: the final original accuracy is slightly higher than the anchor. Fashion-MNIST is essentially unchanged on the original distribution, although the rotated distributions are lower in absolute accuracy.

The relationship between original and shifted performance is important for interpreting the negative result. If the model had oversmoothed the original decision structure in order to fit the rotated domains, we would expect original accuracy to be the uniquely sacrificed quantity. Instead, the harder cases are the shifted validation sets themselves: MNIST $+20^\circ$ and both Fashion-MNIST rotations have lower final accuracy than the original distribution. This pattern is more consistent with ordinary shift difficulty than with a learned representation that has washed out the original class structure. Final DOG values range from 0.0123 to 0.0269, indicating that the average drop from each previous distribution's best point is only about 1–3 percentage points.

We use confusion matrices as qualitative error-structure checks. Figure 1 shows the most diagnostic case: sklearn digits, which has the largest original-domain drop. Errors remain concentrated around a dark diagonal rather than spreading broadly across rows or columns. This is diagnostically important: severe oversmoothing would plausibly appear not only as a small scalar accuracy loss, but also as a diffuse confusion pattern in which class-specific boundaries lose selectivity across many labels.

The observed matrices instead suggest that the learned classifier retains label-specific structure and makes localized mistakes under rotation. Additional MNIST matrices in the appendix show the same diagonal-dominance pattern.

The representation diagnostic is similarly inconclusive for oversmoothing. The final centroid ratios are 2.96 for sklearn digits, 2.22 for MNIST, and 2.63 for Fashion-MNIST. These are slightly below the best observed ratios in each run, but not suggestive of a representation collapse. Together with the confusion matrices, this suggests that the model's class geometry is mildly perturbed by the shifted distributions rather than flattened. This interpretation is qualitative: without multi-seed intervals, CKA, or boundary-local probes, we cannot rule out smaller geometry changes, and we do not claim that these benchmarks are a hard test of distributional oversmoothing.

Several planned ablations, including no replay, architecture changes, padding changes, and weight-decay variants, produced plotting artifacts with empty axes in the available final figure set. Because the corresponding curves are not interpretable from the saved plots, we do not claim evidence from those ablations. This is itself a useful lesson: for negative-result papers, missing or malformed diagnostics should be reported as missing rather than converted into narrative support. We also treat all numbers as completed controlled runs rather than statistically powered estimates, since no multi-seed uncertainty intervals are available in the final summaries.

6 Conclusion

We set out to find distributional oversmoothing: a regime where additional shifted data harms earlier distributions by smoothing away discriminative representation geometry. In the reported controlled rotation experiments, we did not find strong evidence for this phenomenon. With robust normalization and 50% replay, original-distribution degradation was small on sklearn digits and absent or negligible on MNIST and Fashion-MNIST. The most concrete pitfall we found was not a new continual-learning failure mode, but a preprocessing issue: near-zero original-domain pixel variance can make rotated inputs numerically unstable.

The negative result narrows the hypothesis. Distributional oversmoothing may require harder natural shifts, larger models, less replay, or settings where distribution-specific cues conflict more directly. Future work should test higher-resolution datasets, domain shifts such as clipart/photo/painting, explicit representation-similarity metrics such as CKA, and boundary-local probes. Until then, our results suggest caution: before attributing old-domain degradation to a new geometric pathology, verify normalization, replay strength, and diagnostic logging.

References

Supplementary Material

A Hyperparameters and Implementation Details

Table 2: Hyperparameters used in the reported runs.

Dataset	Initial epochs	Continual epochs	Batch	LR	Weight decay	Replay
sklearn digits	30	12	128	7×10^{-4}	5×10^{-5}	0.50
MNIST	14	7	256	8×10^{-4}	5×10^{-5}	0.50
Fashion-MNIST	14	7	256	8×10^{-4}	5×10^{-5}	0.50

All runs used a two-hidden-layer MLP with hidden width 128, ReLU activations, AdamW, cross-entropy loss, and gradient clipping with maximum norm 5.0. The distribution order was fixed as original, $+20^\circ$ rotation with noise standard deviation 0.015, and -20° rotation with noise standard deviation 0.030. For normalization, the mean and standard deviation were estimated from the original training distribution, with each standard deviation floored at 0.10 before applying the transform to all distributions. Inputs were flattened after resizing to 8×8 grayscale arrays.

Table 3: Diagnostics and ablations not used as evidence. These entries are included to make clear which planned analyses did not support claims in the paper.

Planned analysis	Reason not used for claims
CKA representation similarity	Planned but not completed in the reported final runs; only the centroid-ratio diagnostic is reported.
Boundary-sharpness probes	Planned but not completed in the reported final runs, so no quantitative boundary-sharpness claim is made.
Replay, padding, architecture, and regularization ablations	The available saved plots for these ablations contained empty-axis artifacts or otherwise uninterpretable curves, so they are treated as missing diagnostics rather than negative or positive evidence.
Multi-seed uncertainty estimates	The final summaries provide point estimates only; the main text therefore avoids claims about statistical significance.

Table 4: Inventory of saved figures that are intentionally not included. This table documents the figure-selection decision rather than adding new evidence.

Saved figure group	Reason for omission
`*_final_accuracy_by_distribution.png`	Duplicates the numerical content of Table 1; the table is more precise for the small differences observed.
`sklearn_digits_rotations_summary.png`, `hf_mnist_rotations_summary.png`, `hf_fashion_mnist_rotations_summary.png`	Retained as run-summary artifacts but not included because the main paper already reports the completed scalar diagnostics and the appendix reports the most informative qualitative confusion matrix not shown in the main text.
`hf_fashion_mnist_rotations_confusion_matrices.png`	Omitted after figure review: it showed the same diagonal-dominance pattern as the retained examples but added little beyond the Fashion-MNIST row of Table 1.
`*_aggregate_metrics.png` and `*_training_validation_curves.png` for ablation runs	Not used for claims because the available ablation plots were not reliable enough to interpret, as summarized in Table 3.

B Additional Figures

After reviewing all saved figures, we retained one additional supplementary figure. The final-accuracy bar charts were removed rather than reproduced because their numerical content is fully contained in Table 1. The run-summary plots and ablation plots were also excluded because they either duplicate reported quantities or contain malformed axes that would make them unreliable evidence.

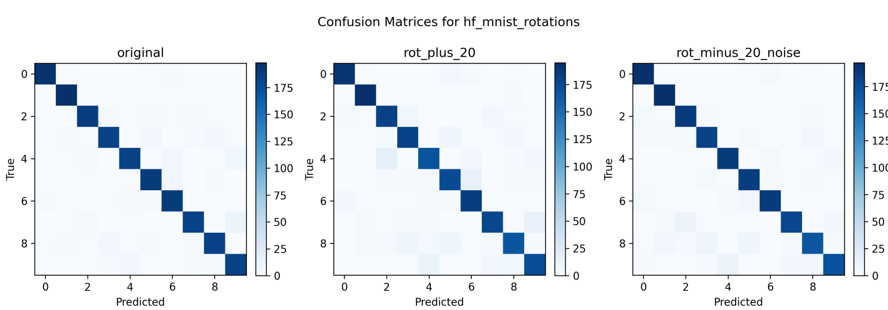

Figure 2: Final MNIST confusion matrices after the complete continual-learning sequence. The original matrix remains diagonal and the shifted matrices show additional but still structured errors, matching the interpretation that rotations are harder inputs rather than evidence of severe oversmoothing. Rows are true classes and columns are predicted classes; darker cells denote larger counts.